Problems (CHAPTER 23) (19, 22, 24, 25, 31, 36, 45, 47)

19- a) $\sigma = ?$
b) $E = ?$ Outside the surface of Satellite due to surface charge.

a) $\sigma = \frac{q}{A}$

$\because$ Diameter of Satellite $= d = 1.3$
$r = \frac{d}{2} \Rightarrow 0.65$
$q = 2.4\mu C$

So,

$$\sigma = \frac{2.4\times10^{-6}}{4\times3.14\times(0.65)^2}$$

$$\boxed{\sigma = 4.5\times10^{-7}\, C/m^2}$$

b) $E = \frac{\sigma}{\epsilon_0} \Rightarrow \frac{4.5\times10^{-7}}{8.85\times10^{-12}} \Rightarrow \boxed{5.1\times10^{4}\, N/C}$

22- $\lambda = 6\,\mu C/m$
$a = ?$

$F = ma$ & $F = qE = eE$

So,

$$ma = eE$$

$$ma = e\frac{\lambda}{2\pi\epsilon_0 r}$$

$$a = \frac{(1.6\times10^{-19})(6\times10^{-6})}{2\times3.14\times8.85\times10^{-12}\times9\times10^{-2}\times9.11\times10^{-31}}$$

$$\boxed{a = 2.1\times10^{17}\, ms^{-2}}$$

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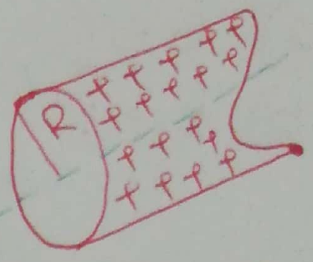
24-

Long thin walled tube

$R = 3cm$

$\lambda = 2\times10^{-8} cm^{-1}$

$E = ?$ at radial distance



$$\begin{cases} r = R/2 < R = \frac{3}{2} = 1.5cm \\ r = 2R \Rightarrow r = 2\times3 = 6cm \end{cases}$$

$$\varphi = \int E \cdot dA$$

$$= \int E(2\pi r) = \frac{q_{enc}}{\varepsilon_0} \quad -(i)$$

a) $r < R$

$$\boxed{E = 0}$$

b) $r > R$

$$E = \frac{\lambda}{2\pi r \epsilon_0} \Rightarrow \frac{2\times10^{-8}}{2(3.14)(8.85\times10^{-12})(6\times10^{-12})}$$

$$\boxed{= 6\times10^{3} N/C}$$

(Matlab Code)

```
lamb = 2x10^-8;
r1 = 0.03;
r2 = 0:0.001:2*r1;
E = lamb./((2*Pi*8.85*10^-12)*r2);
Plot(r2, E);
```

25-

$E = 4.5 \times 10^4 \, N/C$

$d = 2m$

$\lambda = ?$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$\lambda = E(2\pi\epsilon_0 r) \Rightarrow (4.5\times10^4)(2(3.14)(8.85\times10^{-12})(2)$$

$$\boxed{\lambda = 5\times10^{-6} C/m}$$

31-

$r_1 = 3cm$

$r_2 = 6cm$

$\lambda_1 = 5\times10^{-6} C/m$ (inner shell)

$\lambda_2 = -7\times10^{-6} C/m$ (outer shell)

a) $|E| = ?$

b) Direction at radial distance 4cm

c) $E = ?$

d) at $r = 8cm$

Sol

a) $E(r) = \frac{\lambda_1}{2\pi\epsilon_0 r} \Rightarrow \frac{5\times10^{-6}}{2(3.14)(8.85\times10^{-12})(4\times10^{-2})}$

$$\boxed{E = 2.3\times10^{6} N/C}$$

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b) $+E \Rightarrow$ Points radially outward

c) $8cm > 6cm$

$$E(8cm) = \frac{\lambda_1 + \lambda_2}{2\pi\epsilon_0 r} \Rightarrow \frac{(5\times10^{-6})-(7\times10^{-6})}{2(3.14)(8.85\times10^{-12})(8\times10^{-2})}$$

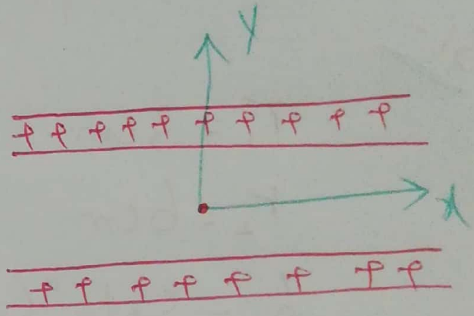
$$\boxed{= -4.5\times10^{5}\ N/C}$$

* $|\lambda_2| > |\lambda_1|$

where (λ_1 is +ve) & (λ_2 is -ve)

36-

$\sigma = 1.77\times10^{-22}\ C/m^2$



a) Unit vector Notation
$\vec{E} = ?$ (Above the sheets)

b) B/w the sheets

c) Below the sheets

Sol

a) $E = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} \Rightarrow \frac{\sigma}{\epsilon_0} \Rightarrow \frac{1.77\times10^{-2}}{8.85\times10^{-12}}$

$$\boxed{= 2\times10^{-11}\ N/C\ \hat{j}}$$

b) Between them

$$\boxed{E=0}$$

c) $\boxed{\vec{E} = (2\times10^{-11})(-\hat{j})\ N/C}$

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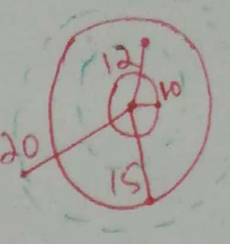
45.

$r_1 = 10cm$ (charged shell 1)

$r_2 = 15cm$ (charged shell 2)



Charge on inner shell $= q_1 = 4\times10^{-8}C$

charge on outer shell $= q_2 = 2\times10^{-8}C$

$E = ?$

a) $r = 12cm$

b) $r = 20cm$

Sol

a) $\vec{E} = \frac{q_1}{4\pi\epsilon_0 r^2} = \frac{(4\times10^{-8})(9\times10^{9})}{(12\times10^{-2})^2}$

$\boxed{\vec{E} = 2.5\times10^{4} N/C}$

b) $\vec{E} = K\frac{q_1+q_2}{r^2}$

$= \frac{(9\times10^{9})((4\times10^{-8})+(2\times10^{-8}))}{(20\times10^{-2})^2}$

$\boxed{\vec{E} = 1.35\times10^{4} N/C}$

47- $d = 15cm$

$r = 10cm$

$E = 3 \times 10^3 \, N/C$

$q_{net} = ?$

$$E = K\frac{q}{r^2}$$

$$q = \frac{Er^2}{K} \Rightarrow \frac{(3\times10^3)(15\times10^{-2})^2}{(9\times10^9)}$$

$$\boxed{q = 7.5\times10^{-9}C}$$

$$\boxed{q = 7.5nC}$$

It should be $\boxed{q = -7.5nC}$ (Because field is radially inward)